Multivariate Calculus Lecture Notes
Section 16.1 – Vector Fields

I. Introduction
   a. We've seen vector-valued functions, functions that assign a vector to a real number.
   b. A 2D *vector field* is a function that assigns a vector to a point in the plane.
   c. Example: $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$.
      i. Note that $\mathbf{F}$ takes two inputs, and the values of $\mathbf{F}$ are vectors.
      ii. Graphically:

      iii. This vector field is sometimes expressed as $\mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$.
   d. General equation for a vector field in the plane: $\mathbf{F}(x, y) = F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j}$, where F_1 and F_2 are functions of x and y.
   e. Physical examples of vector fields
      i. Gravitational fields
      ii. Electric force fields
      iii. Velocity fields
   f. You can have 3D vector fields, too, but they're hard to draw.

II. More examples (perhaps describe the fields as velocity fields?)
   a. $\mathbf{F}(x, y) = x\mathbf{i} - y\mathbf{j}$

   b. $\mathbf{F}(x, y) = \dfrac{y\mathbf{i} - x\mathbf{j}}{\sqrt{x^2 + y^2}}$.

III. Have students find equations for several vector fields via transparencies.

IV. Operations on vector fields
   a. Recap of gradients
      i. Recall that the gradient of $f(x, y, z)$ is given by $\nabla f(x, y, z) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$.
      ii. The symbol ∇ is called "del," and it is sometimes thought of as an operator: $\nabla = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle$.
      iii. Thus ∇f takes the partials of f and places them into a vector.
   b. Divergence

i. The *divergence* of a vector field $\mathbf{F} = \langle F_1, F_2, F_3 \rangle$ is given by

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

ii. Example: div $\mathbf{F}(-2, 0, 5)$, where $\mathbf{F}(x, y, z) = xz\mathbf{i} + xyz\mathbf{j} - y^2\mathbf{k}$.
   1. We have div $\mathbf{F} = z + xz$.
   2. Thus div $\mathbf{F}(-2, 0, 5) = 5 - 10 = -5$.

iii. Linearity properties
   1. $\text{div}(\mathbf{F} + \mathbf{G}) = \text{div } \mathbf{F} + \text{div } \mathbf{G}$
   2. $\text{div}(c\mathbf{F}) = c \text{ div } \mathbf{F}$

iv. Interpretation
   1. Generally, divergence is related to how vectors in a vector field move across a point.
   2. If div $\mathbf{F}(x, y, z) > 0$, then (x, y, z) is a *source*; vectors in the field tend to point away from it.
   3. If div $\mathbf{F}(x, y, z) < 0$, then (x, y, z) is a *sink*; vectors in the field tend to point towards it.
   4. If div $\mathbf{F}(x, y, z) > 0$, then (x, y, z) is a *incompressible*; vectors in the field tend to point across it.

v. Discuss source/sink/incompressible points on transparencies.

c. Curl

i. The *curl* of a 3D vector field $\mathbf{F} = \langle F_1, F_2, F_3 \rangle$ is given by

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} - \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}.$$

ii. The *curl* of a 2D vector field $\mathbf{F} = \langle F_1, F_2 \rangle$ is given by

$$\text{curl } \mathbf{F} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}.$$

iii. Example: Find curl $\mathbf{F}(-2, 0, 5)$, where $\mathbf{F}(x, y, z) = xz\mathbf{i} + xyz\mathbf{j} - y^2\mathbf{k}$.
   1. We have curl $\mathbf{F} = (-2y - xy)\mathbf{i} - (0 - x)\mathbf{j} + (yz - 0)\mathbf{k} = (-2y - xy)\mathbf{i} + x\mathbf{j} + yz\mathbf{k}$.
   2. Thus curl $\mathbf{F}(-2, 0, 5) = -2\mathbf{j}$.

iv. Linearity properties
   1. $\text{curl}(\mathbf{F} + \mathbf{G}) = \text{curl } \mathbf{F} + \text{curl } \mathbf{G}$
   2. $\text{curl}(c\mathbf{F}) = c \text{ curl } \mathbf{F}$

v. Interpretation
   1. If we consider $\mathbf{F}$ to be the velocity vector field for a fluid flowing through a region, then the curl at the point P contains information about how fast and in which direction a small paddle wheel will spin if it is placed at P.
      - If the paddle wheel is placed perpendicular to curl $\mathbf{F}$ then it will spin the fastest.
      - The magnitude of curl $\mathbf{F}$ is the speed at which the paddle wheel would rotate when placed thusly.
   2. In general, curl is related to the tendency of a vector field to rotate about a point.

vi. Discuss curl of vector fields from transparencies

V. Conservative vector fields

a. A vector field $\mathbf{F}$ is called *conservative* if there exists some function V such that $\mathbf{F} = \nabla V$.
   i. If V exists, it is called the *potential function* for $\mathbf{F}$.
   ii. The term "conservative" comes from the concept of conservation of energy.
   iii. For example, gravitational fields and electric fields are conservative, but magnetic ones are not.

b. Examples
   i. If $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$ a conservative vector field?
      1. Well, is there some function V such that $V_x(x, y) = x$ and $V_y(x, y) = y$?
      2. If there were such a function it would have to have the following form:
         - $V(x, y) = \frac{1}{2}x^2 + g(y)$ (from $V_x(x, y) = x$)
         - $V(x, y) = \frac{1}{2}y^2 + h(x)$ (from $V_y(x, y) = y$)
         - So we have $V(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2 + K$ for some constant K.
      3. Thus $\mathbf{F}$ is a conservative vector field.
      4. This example illustrates the "uniqueness of potential functions" property: If $\mathbf{F}$ is conservative on an open connected domain, then any two potential functions $\mathbf{F}$ differ by a constant.
   ii. Is $\mathbf{F}(x, y) = y\mathbf{i}$ a conservative vector field?
      1. Try to integrate:
         - $V(x, y) = xy + g(y)$
         - $V(x, y) = h(x)$
      2. There's no way we can create one function V with these conditions.
      3. Thus $\mathbf{F}$ is not a conservative vector field.

c. Testing for conservative vector fields
   i. Trying to find the potential function in 3D is a bit more difficult than in 2D.
   ii. There is a straightforward test, though, that works in 2D and in 3D that doesn't calculate the potential function.
      1. Suppose we have $\mathbf{F}(x, y) = F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j}$.
      2. If $\mathbf{F}$ really is a conservative vector field, then there is some potential function $V(x, y)$ such that $V_x(x, y) = F_1(x, y)$ and $V_y(x, y) = F_2(x, y)$.
      3. Now, we know that, except in some really oddball cases, the mixed second partial derivatives of V should be equal.
      4. So if V really does exist we should have $\dfrac{\partial F_1}{\partial y} = \dfrac{\partial F_2}{\partial x}$.
   iii. Theorem: If $\mathbf{F}(x, y) = F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j}$, and F_1 and F_2 have continuous first partial derivatives in some open simply-connected region R, then $\mathbf{F}$ is a conservative vector field in R if and only if $\dfrac{\partial F_1}{\partial y} = \dfrac{\partial F_2}{\partial x}$.
   iv. The 3D case, with $\mathbf{F}(x, y, z) = F_1(x, y, z)\mathbf{i} + F_2(x, y, z)\mathbf{j} + F_3(x, y, z)\mathbf{k}$, works similarly.
      1. You have to check all possible mixed partial derivatives.
      2. Thus $\mathbf{F}$ is conservative in this situation if and only if $\dfrac{\partial F_1}{\partial y} = \dfrac{\partial F_2}{\partial x}$, $\dfrac{\partial F_1}{\partial z} = \dfrac{\partial F_3}{\partial x}$, and $\dfrac{\partial F_2}{\partial z} = \dfrac{\partial F_3}{\partial y}$.
      3. In other words, $\mathbf{F}$ is conservative iff curl $\mathbf{F} = \mathbf{0}$.
   v. Is $\mathbf{F} = y\mathbf{i} + 2\mathbf{j} + x\mathbf{k}$ a conservative vector field?

1. We have $\dfrac{\partial F_1}{\partial y} = 1$ and $\dfrac{\partial F_2}{\partial x} = 0$.
2. Thus $\mathbf{F}$ is not conservative.

Homework: Section 16.1 #2, 8, 11, 13-16, 23, 26, 30, 43, 44, 47, 50, 52.